Slat control and monitoring system

Training — line maintenance familiarization

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Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Training Centre Valmont (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	MRDX-42 and MRDX-46, all aircraft		

1. Lesson plan

Target audience: B1 licensed engineers converting to the MRDX. Duration: 45 minutes, classroom, followed by 30 minutes on the maintenance trainer.

Objective 1:
State the function of the two slat flap control computers and describe how control passes from the master to the standby computer.

Objective 2:
Identify the power control unit, the feedback position pick-off units and the wing tip brakes on the aircraft, and state the access panel for each.

Objective 3:
Interpret the fault message SLAT SYS 1 FAULT and select the correct fault isolation data module.

2. Overview

The slat system moves the leading edge surfaces between five positions. It is a two-channel system: each channel has its own computer, its own hydraulic motor and its own position feedback. A failure of one channel reduces the rate of movement but does not remove the function.

The part of the system that most often causes a line maintenance event is the position feedback. Moisture in the bearing housing of a feedback position pick-off unit causes the signal to drift, which the monitoring function detects before the surface position is affected.

3. Content

1 Control path

The lever position is read by two independent sensors. The master computer calculates the commanded position, compares it with the feedback and drives the valve block of the power control unit until the difference is inside the tolerance band.

2 Monitoring

The asymmetry position pick-off units at the outboard end of each transmission are compared continuously. A difference of more than 3.6 degrees engages the wing tip brakes and latches the system until it is reset on the ground.

NOTE: A latched system cannot be reset in flight. The flight crew drill for a jammed slat applies for the rest of the flight.

4. Assessment

Question 1

Which condition engages the wing tip brakes?

A.
A difference of more than 1.0 degree between the commanded and the actual position.

B. □
A difference of more than 3.6 degrees between the left and the right asymmetry position pick-off units.

C.
Loss of one hydraulic system.

Question 2

The fault message SLAT SYS 1 FAULT is shown with the correlated message SFCC1. What is the first action of the fault isolation procedure?

A. □
Do the SFCC 1 built-in test on the maintenance terminal.

B.
Replace the SFCC 1.

C.
Replace both feedback position pick-off units.

5. Summary

The slat system is tolerant of a single channel failure but not of an undetected asymmetry. Confirm the fault with the built-in test before any unit is removed.